

Probability



Excel Mathematics Study Guide Years 9–10
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UNIT 4: Independent events

QUESTION 1 Complete:

Two events are independent if the outcome of the first event does _____ the outcome of the second event.

QUESTION 2 Determine whether the pair of events are dependent or independent:

- a tossing two coins _____
- b tossing a coin and throwing a die _____
- c taking and eating 2 jellybeans from a bowl, one after the other _____
- d selecting two cards from a pack, one after the other, with replacement _____
- e selecting two cards from a pack, one after the other, without replacement _____

QUESTION 3 A die is rolled and a coin is tossed. What is the probability of getting:

- a a head on the coin? _____
- b a 4 on the die? _____
- c a 4 and a head? _____

QUESTION 4 A coin is tossed three times. What is the probability that:

- a the first toss is a head? _____
- b the second toss is a head? _____
- c the third toss is a head? _____
- d all three tosses are heads? _____

QUESTION 5 A bag holds 5 red, 3 blue and 2 green counters. A counter is selected at random, its colour noted, and it is then replaced. A second counter is then selected at random. What is the probability that:

- a the first counter is red? _____
- b the second counter is red? _____
- c both counters are red? _____
- d both counters are blue? _____
- e both counters are green? _____
- f one is red and one is blue? _____
- g neither is green? _____
- h at least one is green? _____

QUESTION 6 A basket holds 6 white, 4 black and 2 grey pegs. Three pegs are taken, one after the other with replacement. What is the probability that:

- a all the pegs are white? _____
- b all the pegs are black? _____
- c all the pegs are grey? _____
- d none are black? _____
- e at least one is black? _____
- f at least one is grey? _____

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UNIT 5: Dependent events

QUESTION 1 Complete:

- a Two events are dependent if the outcome of the first event _____ the outcome of the second event.

QUESTION 2 Determine whether the pair of events are dependent or independent:

- a throwing two dice _____
b spinning a spinner twice _____
c winning two prizes in a raffle _____
d taking two marbles from a bag, one after the other, without replacement _____
e taking two marbles from a bag, one after the other, with replacement _____

QUESTION 3 Jesse buys 5 tickets in a raffle. 1000 tickets are sold altogether. There are two prizes. A ticket is drawn for first prize and this ticket is discarded before a second ticket is drawn. What is the probability that Jesse wins:

- a first prize? _____
b second prize if he didn't win first prize? _____
c second prize if he did win first prize? _____
d both prizes? _____

QUESTION 4 A bag holds 5 red, 3 blue and 2 green counters. A counter is selected at random and is not replaced. A second counter is then selected at random. What is the probability that:

- | | |
|-----------------------------------|---|
| a the first counter is red? _____ | b the second counter is also red? _____ |
| c both counters are red? _____ | d both counters are blue? _____ |
| e both counters are green? _____ | f one is red and one is blue? _____ |
| g neither is green? _____ | h at least one is green? _____ |

QUESTION 5 A basket holds 6 white, 4 black and 2 grey pegs. Three pegs are taken, one after the other without replacement. What is the probability that:

- | | | |
|---|---|--|
| a all the pegs are white?

_____ | b all the pegs are black?

_____ | c all the pegs are grey?

_____ |
| d none are black?

_____ | e at least one is black?

_____ | f at least one is grey?

_____ |